

Unit 4, Lesson 11: Writing Equations of Perpendicular Lines



Benchmark

MA.912.AR.2.3 Write a linear two-variable equation for a line that is parallel or perpendicular to a given line and goes through a given point.



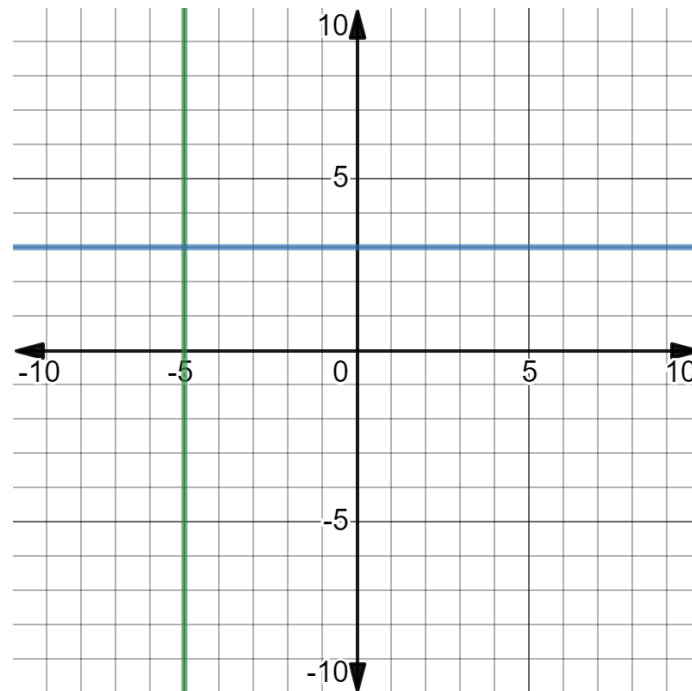
Learning Targets

- ☐ I can write the equation of a line that is perpendicular to a given line through a given point.
- ☐ I can write the equation of a line that is perpendicular to a given line when given two points.
- ☐ I can write the equation of a line that is perpendicular to a given line when given a point and the slope.



4.11.1 Warm-Up: Horizontal and Vertical Lines

The lines $x = -5$ and $y = 3$ are shown on the coordinate grid.



Rodrigo completed the table for the lines $x = -5$ and $y = 3$.

Line	Slope	Explanation
$x = -5$	$\frac{2}{0}$	If I start at the point $(-5, 0)$, I can go up 2 and still be on the line. In terms of $\frac{\text{rise}}{\text{run}}$, I rise 2 and run 0.
$y = 3$	$\frac{0}{2}$	If I start at the point $(0, 3)$, I can go up 0 and over 2 and still be on the line. In terms of $\frac{\text{rise}}{\text{run}}$, I rise 0 and run 2.

1. Give another slope Rodrigo could have used for each line.

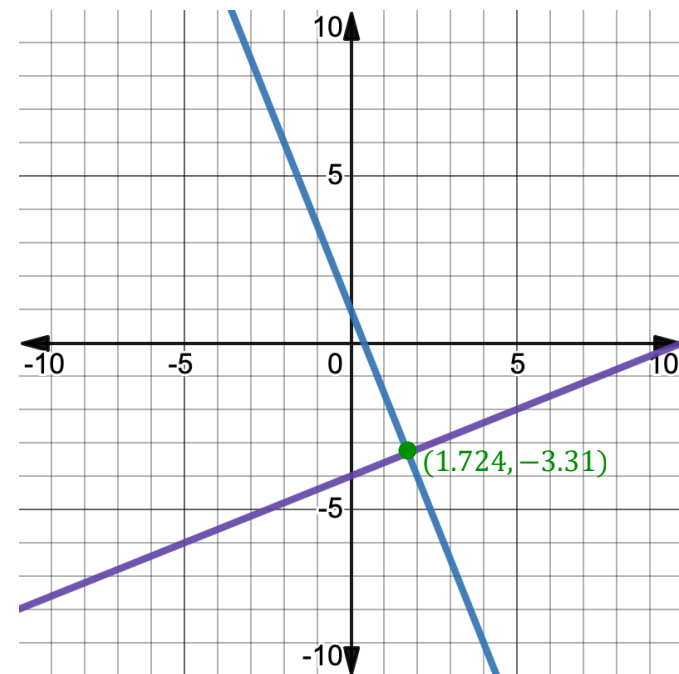
Line	Slope
$x = -5$	
$y = 3$	

2. Write the equation of the line that passes through $(2, 1)$ and is parallel to $x = -5$.
3. Write the equation of the line that passes through $(2, 1)$ and is parallel to $y = 3$.
4. What is the point where $x = -5$ and $y = 3$ intersect?



4.11.2 Guided Instruction: Writing Equations of Perpendicular Lines Through a Point

1. In the Warm-Up, what is the relationship between the lines $x = -5$ and $y = 3$?
2. Two lines are graphed on the coordinate grid. The lines are perpendicular to each other.



- a. Find the slope of each line.

Line	Slope
Purple	
Blue	

- b. Write the equation of the purple line.
- c. Write the equation of the blue line.
- d. Discuss with your partner how to write the equation of a line that is perpendicular to another line. Write notes from the discussion.



Lines are **perpendicular** if the slopes are opposites and reciprocals.

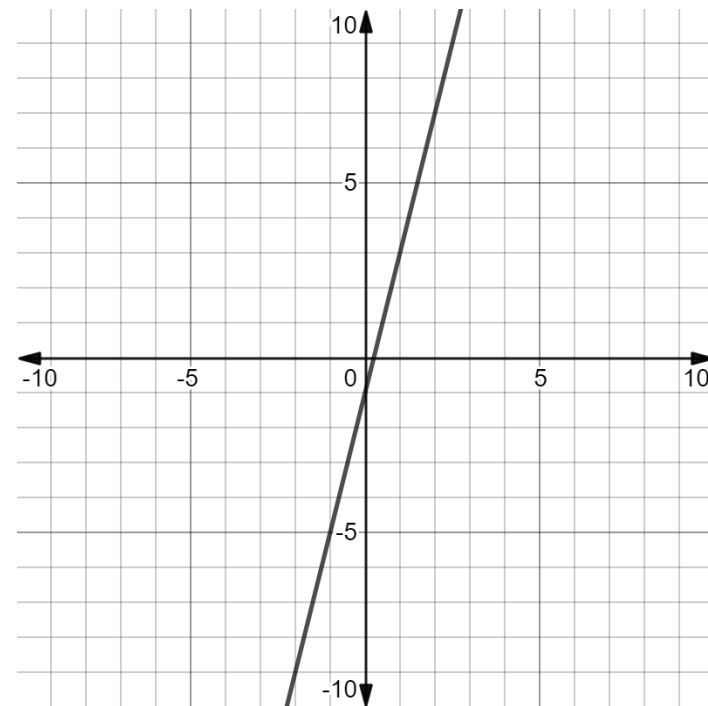
3. Write the equation of the line, in all three forms, that is perpendicular to the line that passes through the points $(-5, -2)$ and $(3, 2)$ and goes through the point $(4, -1)$.

Point-Slope Form	Slope-Intercept Form	Standard Form



4.11.3 Guided Practice: Writing Equations of Perpendicular Lines Through a Point

1. Consider the following graph.



- a. Write the equation of the line that is perpendicular to the line shown on the coordinate grid and passes through the point $(5, 2)$.
- b. What is the point where the two lines intersect?

- c. Write the equation of the line, in all three forms, that is perpendicular to the line $25x - 5y = 45$ and goes through the point $(0, 7)$.

Point-Slope Form	Slope-Intercept Form	Standard Form

2. Write the equation of the line, in all three forms, that is perpendicular to the line $y - 8 = \frac{5}{6}(x + 11)$ and goes through the point, $(-10, -3)$.

Point-Slope Form	Slope-Intercept Form	Standard Form



4.11.4 Your Turn!

- Write the equation of the line that goes through the point $(6, -8)$ and is perpendicular to the line that passes through the points $(-7, -12)$ and $(-7, 6)$.
- Write the equation of the line that goes through the point $(2, 3)$ and is perpendicular to the line $-8x - 6y = -48$.



4.11.5 Wrap-Up: Reflection

1. Complete the learning pyramid.

One question I would like
to have answered ...

One mistake I learned
from today ...

One thing I am not sure
about ... YET!

One thing I am confident I know is ...

Unit 4, Lesson 12: Writing Equations of Parallel and Perpendicular Lines



Benchmark

MA.912.AR.2.3 Write a linear two-variable equation for a line that is parallel or perpendicular to a given line and goes through a given point.



Learning Targets

- ☐ I can write an equation for a line parallel to a given line through a given point.
- ☐ I can write an equation for a line perpendicular to a given line through a given point.